

VINODH BALAKRISHNA

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PROFESSIONAL SUMMARY

Embedded Systems Software Engineer with deep expertise in low-level programming, OS internals, and Linux-based platforms. Experienced in firmware development, hardware bring-up, and debugging complex system-level issues across ARM and x86 architectures. Skilled at writing clean, efficient code and bridging the hardware-software boundary. A focused problem-solver with a curious mindset, thriving both independently and in collaborative, fast-paced environments to deliver reliable, maintainable solutions.

EXPERIENCE

Embedded Linux Engineer

Chennai, India

Comcast Corporation

Dec 2021 - Jul 2023

- Contributed to the Platform Security team by leveraging the Yocto Project build system and RDK software stack to create and manage custom Linux firmware for over 1 million CPE devices, including set-top boxes, routers, and TVs.
- Secured and optimized a custom Restricted Shell with a multi-threaded architecture by implementing command logging, unauthorized command blocking, and tab completion, boosting efficiency and usability by 40%.
- Migrated SBOM functionality from Yocto Kirkstone to Dunfell to enhance supply chain security and maintain 100% software inventory transparency.
- Renewed and rotated 5+ X.509 PKI certificates, ensuring uninterrupted secure TLS communication with zero certificate expiry incidents during release cycles.
- Performed key platform security enhancements to RDKB devices by enabling ASLR with position-independent code and securely removing video analytics module, improving firmware performance by 25% while reducing maintenance overhead.
- Accessed Linux-based CPE devices on ARM and Atom platforms for board/device bring-up and firmware debugging by connecting via Raspberry Pi serial console using Minicom with appropriate baud rate settings.
- Used CI/CD pipelines with Jenkins and Shell scripting to trigger and monitor automated builds and tests in an Agile environment, while utilizing Gerrit for code commits and reviews.

Programmer Analyst Trainee (Intern)

Chennai, India

Cognizant Solutions

Aug 2021 - Dec 2021

- Led a PoC to integrate ShellCheck for automated shell script error detection, reducing script errors by 30%.
- Addressed 10+ critical CVE vulnerabilities by applying timely security patches, enhancing overall software security.
- Transitioned over 1000 lines of standard C code to Safe C library, significantly decreasing memory-related vulnerabilities.

TECHNICAL PROJECTS

Torrent CLI - BitTorrent Client | Source Code: https://github.com/v1n0dh/torrent_cli

- Developed a high-performance BitTorrent client in modern C++23, enabling efficient concurrent downloads and full support for the BitTorrent protocol including peer communication, tracker requests, and piece/block management.
- Implemented zero-copy memory-mapped I/O using `mmap` and `RAII`, thread-safe message passing with custom `Shared_Queue` classes, and concurrency control using mutexes and futures to coordinate peer threads.
- Leveraged ASIO library for scalable asynchronous networking to handle multiple peer connections with fine-grained control, and designed full Bencode encoder/decoder for .torrent metadata serialization and deserialization.
- Used debugging and analysis tools including Wireshark for network traffic inspection, Valgrind for memory diagnostics, and strace to trace syscalls and debug low-level I/O behavior.

Custom Linux Shell (Vshell) | Source Code: <https://github.com/v1n0dh/vshell>

- Built a custom shell in C++, inspired by a coding challenge at codingchallenges.fyi, closely resembling bash, completing the project within 2 weeks to demonstrate core shell functionalities.
- Incorporated key shell features, including command execution, I/O redirection, and pipes, facilitating efficient user interaction with the system.
- Integrated tab completion for files and command history, enhancing user experience and productivity.

Real-Time System Metrics Statusbar | Source Code: <https://github.com/v1n0dh/dwmstatus>

- Created a custom status bar for the dwm window manager in C that displays real-time system metrics such as network speeds, Wi-Fi SSID, CPU temperature, RAM usage, battery status, and current time.
- Collected system data by directly reading low-level kernel interfaces under `/sys/class` for power supply, thermal zones, memory, and network statistics, and used X11 library to dynamically update the status bar on the dwm.
- Implemented network speed calculation by tracking changes in RX/TX bytes per second from network interface statistics, and used wireless extensions `ioctl` calls to fetch Wi-Fi SSID, providing live connectivity information.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Shell Scripting, Java, Rust, x86 Assembly

Core Competencies: Data Structures & Algorithms, Memory Management, Concurrency (Multi-threading/Processing), IPC

Operating Systems & Embedded: GNU Linux/Unix Systems, Embedded Linux Firmware, RTOS

Networking & Communication: TCP/IP, UDP, SNMP, HTTP, TLS/SSL, Networking Protocols, UART, SPI, I2C

Frameworks & Libraries: Yocto Project, RDK, GNU Readline, Jsoncpp, ASIO (non-Boost)

Tools: CMake, GDB, QEMU, OpenSSL, Wireshark, Jenkins, Git, Splunk, Vim, Emacs

EDUCATION

University of Central Missouri

Master's in Computer Science

Lee's Summit, MO

Aug 2023 - May 2025

Coursework: Operating Systems, Algorithms, Computer Networking & Security, Compiler Design & Construction, Cryptographic Algorithms and Protocols, Web Application Security, Cloud Security

Bharath Institute of Higher Education and Research

B.Tech in Computer Science Engineering

Chennai, India

Jul 2017 - May 2021

CERTIFICATIONS

- Rust Essential Training by Barron Stone, LinkedIn Learning - [Certificate Link](#)
- Linux Device Drivers by Kevin Dankwardt, LinkedIn Learning - [Certificate Link](#)